Digital Signal Processing

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Symmetry in FFT Algorithm

- Symmetry in Time Extraction FFT Algorithm
- Symmetry in Frequency Extraction FFT Algorithm
- Symmetry between time extraction and frequency extraction

Symmetry in FFT Algorithm

The coefficient matrix of time-domain to frequency-domain transformation is the same as the coefficient matrix of frequency-domain synthesis

Time domain to frequency domain

$$\begin{bmatrix} X[0] \\ X[1] \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} x[0] \\ x[1] \end{bmatrix}$$

Frequency domain synthesis

$$\begin{bmatrix} X[m] \\ X[m+N/2] \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} W_N^0 & 0 \\ 0 & W_N^m \end{bmatrix} \begin{bmatrix} X_1[m] \\ X_2[m] \end{bmatrix}$$

Base 2 time extraction FFT

Symmetry in FFT Algorithm

The coefficient matrix of time-domain to frequency-domain transformation is the same as the coefficient matrix of frequency-domain synthesis

Time domain to frequency domain

$$\begin{bmatrix} X[0] \\ X[1] \\ X[2] \\ X[3] \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix} \begin{bmatrix} x[0] \\ x[1] \\ x[2] \\ x[3] \end{bmatrix}$$

Frequency domain synthesis

$$\begin{bmatrix} X[m] \\ X[m+N/4] \\ X[m+2N/4] \\ X[m+3N/4] \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix} \begin{bmatrix} W_N^0 & 0 & 0 & 0 \\ 0 & W_N^m & 0 & 0 \\ 0 & 0 & W_N^{2m} & 0 \\ 0 & 0 & 0 & W_N^{3m} \end{bmatrix} \begin{bmatrix} X_1[m] \\ X_2[m] \\ X_3[m] \\ X_4[m] \end{bmatrix}$$

Base 4 time extraction FFT

Symmetry in Frequency Extraction FFT Algorithm

The coefficient matrix of time-domain decomposition is the same as the coefficient matrix of time-domain to frequency-domain transformation

Time-domain decomposition

$$\begin{bmatrix} x_1[k] \\ x_2[k] \end{bmatrix} = \begin{bmatrix} W_N^0 & 0 \\ 0 & W_N^k \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} x[k] \\ x[k+N/2] \end{bmatrix}$$

Time domain to frequency domain

$$\begin{bmatrix} X[0] \\ X[1] \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} x[0] \\ x[1] \end{bmatrix}$$

Base 2 frequency extraction FFT

Symmetry in Frequency Extraction FFT Algorithm

The coefficient matrix of time-domain decomposition is the same as the coefficient matrix of time-domain to frequency-domain transformation

Time-domain decomposition

$$\begin{bmatrix} x_1[k] \\ x_2[k] \\ x_3[k] \\ x_4[k] \end{bmatrix} = \begin{bmatrix} W_N^0 & 0 & 0 & 0 \\ 0 & W_N^k & 0 & 0 \\ 0 & 0 & W_N^{2k} & 0 \\ 0 & 0 & 0 & W_N^{3k} \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -\mathrm{j} & -1 & \mathrm{j} \\ 1 & -1 & 1 & -1 \\ 1 & \mathrm{j} & -1 & -\mathrm{j} \end{bmatrix} \begin{bmatrix} x[k] \\ x[k+N/4] \\ x[k+2N/4] \\ x[k+3N/4] \end{bmatrix}$$

Time domain to frequency domain

$$\begin{bmatrix} X[0] \\ X[1] \\ X[2] \\ X[3] \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -\mathrm{j} & -1 & \mathrm{j} \\ 1 & -1 & 1 & -1 \\ 1 & \mathrm{j} & -1 & -\mathrm{j} \end{bmatrix} \begin{bmatrix} x[0] \\ x[1] \\ x[2] \\ x[3] \end{bmatrix}$$

Base 4 frequency extraction FFT

The coefficient matrix synthesized in the frequency domain of the FFT algorithm based on R-time extraction is the same as the coefficient matrix decomposed in the time domain of the FFT algorithm based on *R*-frequency extraction, and the form of the rotation factor matrix is the same.

Base 2 time extraction FFT Frequency domain synthesis

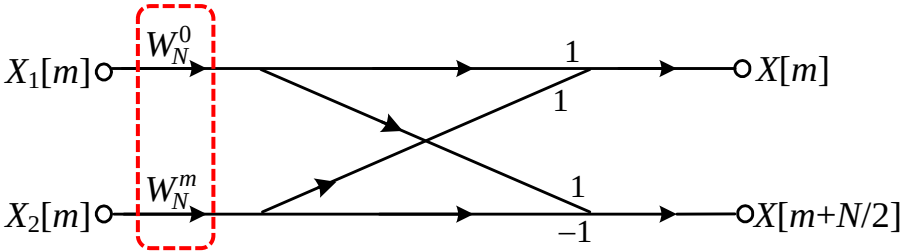
$$\begin{bmatrix} X[m] \\ X[m+N/2] \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} W_N^0 & 0 \\ 0 & W_N^m \end{bmatrix} \begin{bmatrix} X_1[m] \\ X_2[m] \end{bmatrix}$$

Base 2 frequency extraction FFT Time-domain decomposition

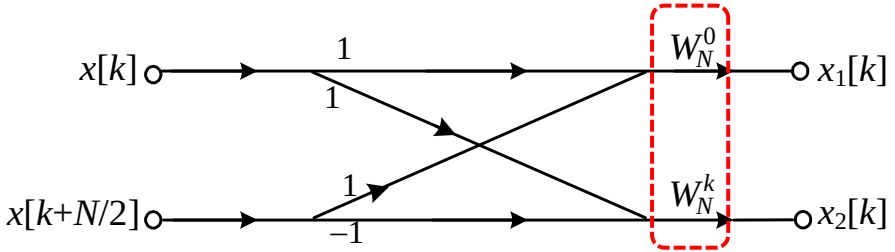
$$\begin{bmatrix} x_1[k] \\ x_2[k] \end{bmatrix} = \begin{bmatrix} W_N^0 & 0 \\ 0 & W_N^k \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} x[k] \\ x[k+N/2] \end{bmatrix}$$

The coefficient matrix synthesized in the frequency domain of the FFT algorithm based on *R*-time extraction is the same as the coefficient matrix decomposed in the time domain of the FFT algorithm based on R-frequency extraction, and the form of the rotation factor matrix is the same.

Base 2 time extraction FFT Frequency domain synthesis



Base 2 frequency extraction FFT Time-domain decomposition



Symmetry between time extraction and frequency extraction

The coefficient matrix synthesized in the frequency domain of the FFT algorithm based on R-time extraction is the same as the coefficient matrix decomposed in the time domain of the FFT algorithm based on R-frequency extraction, and the form of the rotation factor matrix is the same.

Base 4 time extraction FFT Frequency domain synthesis

$$\begin{bmatrix} X[m] \\ X[m+N/4] \\ X[m+2N/4] \\ X[m+3N/4] \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -\mathrm{j} & -1 & \mathrm{j} \\ 1 & -1 & 1 & -1 \\ 1 & \mathrm{j} & -1 & -\mathrm{j} \end{bmatrix} \begin{bmatrix} W_N^0 & 0 & 0 & 0 \\ 0 & W_N^m & 0 & 0 \\ 0 & 0 & W_N^{2m} & 0 \\ 0 & 0 & 0 & W_N^{3m} \end{bmatrix} \begin{bmatrix} X_1[m] \\ X_2[m] \\ X_3[m] \\ X_4[m] \end{bmatrix}$$

Base 4 frequency extraction FFT Time-domain decomposition

$$\begin{bmatrix} x_1[k] \\ x_2[k] \\ x_3[k] \\ x_4[k] \end{bmatrix} = \begin{bmatrix} W_N^0 & 0 & 0 & 0 \\ 0 & W_N^k & 0 & 0 \\ 0 & 0 & W_N^{2k} & 0 \\ 0 & 0 & 0 & W_N^{3k} \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -\mathrm{j} & -1 & \mathrm{j} \\ 1 & -1 & 1 & -1 \\ 1 & \mathrm{j} & -1 & -\mathrm{j} \end{bmatrix} \begin{bmatrix} x[k] \\ x[k+N/4] \\ x[k+2N/4] \\ x[k+3N/4] \end{bmatrix}$$

Thanks

The materials referenced in this course are accumulated over years of teaching by the instructor, sourced from various media and exchanges with colleagues and peers. It is difficult to indicate the source of each material one by one. This is to clarify and express gratitude.